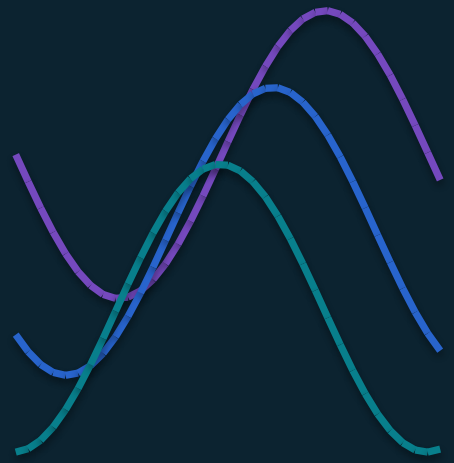
AFS Assessment Assistant

A reviewable first experiment for in-line spectroscopy.



Six final cases. Visible evidence. Explicit uncertainty.

6	FINAL CASES
4	FEATURE DEMOS
16	QUALITY CHECKS

Eisa Atif · Marie Habermann · Atfan Deshmukh · Akshdeep Singh

Naumanurrahman Shaikmujeeburrahman

Six cases test technical decisions.

The challenge is whether the next technical step is justified.

H01 · Calibration transfer

New API blend: revalidate the NIR model.

H02 · Optical geometry

Clear whey and opaque milk need different interfaces.

H03 · Selectivity

Sensitivity alone does not establish selectivity.

H04 · Fiber compatibility

An existing silica link cannot be approved for MIR.

H05 · Undefined target

Clarification needed; Council remains provisional.

H06 · Evidence scope

An in-vitro kidney study does not establish surgical readiness.

JUDGING WEIGHTS



Technical 30% | AFS knowledge 20% | Feasibility 15% | Clarification 15% | Explainability 10% | Prototype 10%

Official final-case inputs H01–H06; challenge rubric p.3

SAVED COUNCIL RESULTS

Useful output, with measurable grounding gaps.

Six archived final answers · three model drafts · automatic checks after generation

35 / 44 source checks verified

4 unsupported-number flags

0 / 6 high-confidence outputs

Case	Source checks	Checks passed	Number flags	Confidence	Decision status
H01	6/8	15/16	0	Low	Provisional
H02	9/10	12/16	1	Insufficient	Provisional
H03	5/5	15/16	0	Low	Provisional
H04	7/10	15/16	0	Insufficient	Provisional
H05	3/3	13/16	3	Insufficient	Provisional
H06	5/8	12/16	0	Insufficient	Outside scope

Automatic checks are not expert accuracy scores. Quote matching covers only selected report fields.

LLMs draft. Code checks the decision.

A structured pipeline keeps sources, assumptions, proposals and uncertainty visible.

01

Read + clarify

Structured requirements and missing facts

02

Rules + retrieval

AFS claims, sources and physical constraints

03

Draft + review

Fast or Council mode; check and repair

04

Gate + explain

Computed confidence; report and Coach

95 curated claim cards

35 sourced rules

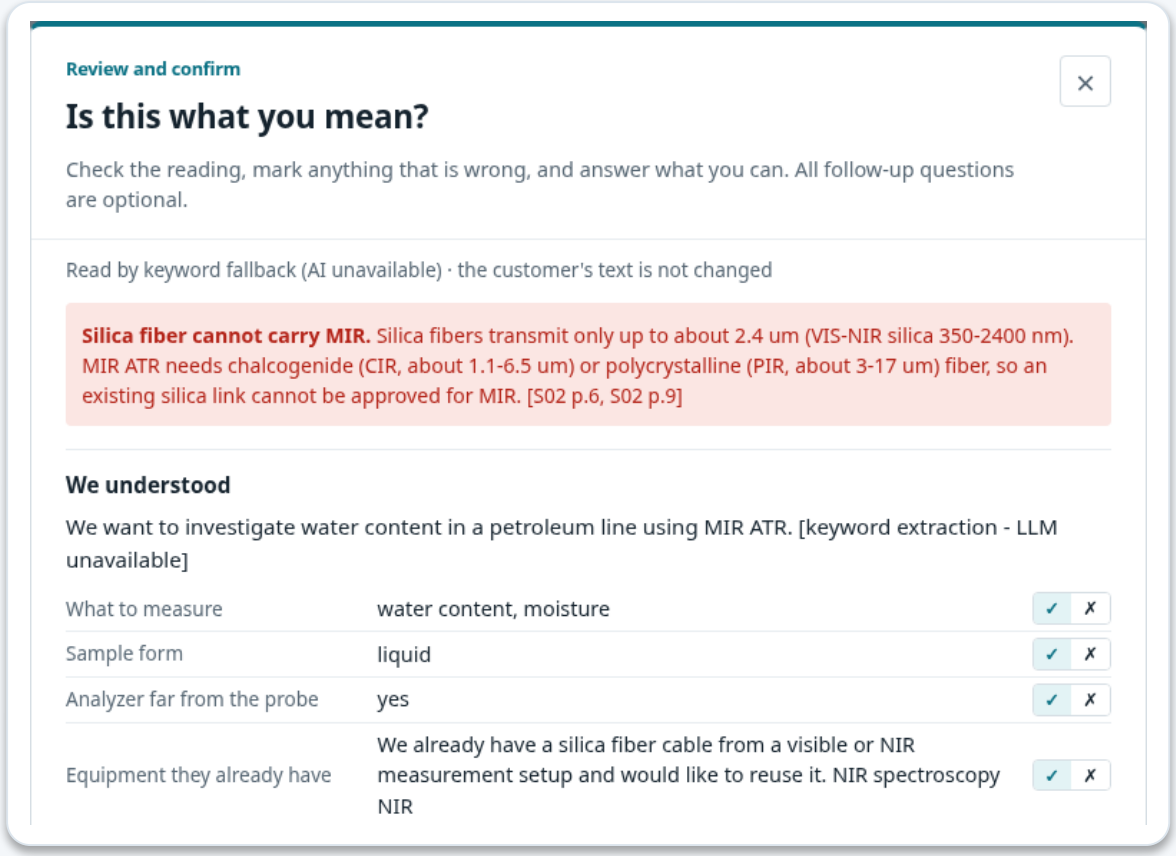
16 quality checks

FastAPI + Pydantic / Groq models / native JavaScript UI / SQLite or Supabase

README; docs/frontend-integration.md; backend/app/engine/{pipeline,verify,quality,confidence}.py

Confirm the reading before assessing.

Show what the assistant understood, then ask what would change the recommendation.



Ask the decision questions

Sample optics

Clear, turbid or opaque — does it change?

Required range

What range and smallest change matter?

Independent reference

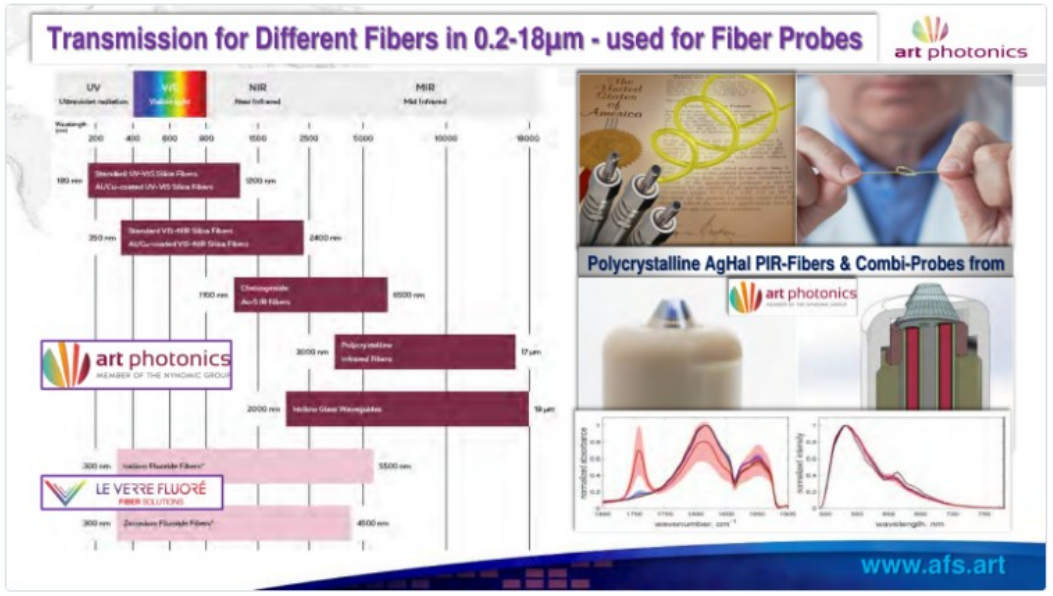
Which reference method provides calibration values?

REVIEW → YES, ASSESS · CLARIFICATION 15%

Open the source that changes the decision.

H04: MIR ATR may be investigated; the existing silica link cannot be approved.

S02 · Page 6 · Line 4 Close



ACTUAL PDF PANE · SOURCE PAGE SHOWN BESIDE THE REPORT

The physical constraint

Silica: VIS / NIR

MIR: a different fiber system

Remote link: confirm with AFS

A lab application is evidence to investigate — not approval of this installation.

Turn the recommendation into a testable comparison.

H04 example · a proposed experiment, not a claim of achieved sensor performance

01 · Samples

Representative petroleum; vary water content and process conditions.

02 · Independent reference

Measure water with a suitable reference, e.g. Karl Fischer.

03 · Compare concepts

NIR on the silica link vs MIR ATR with a confirmed IR link.

04 · Decide with the client

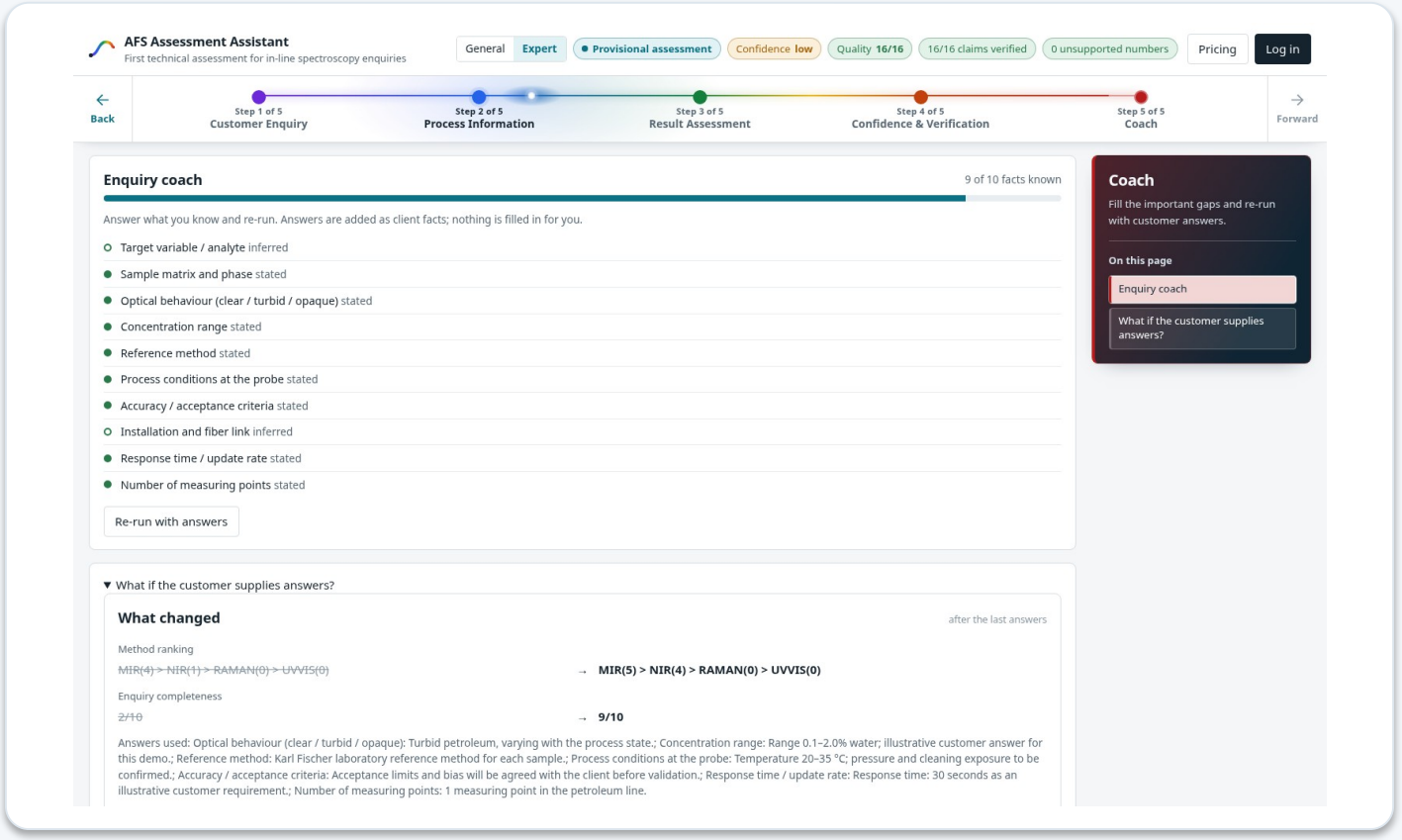
Use held-out samples; judge error, robustness and installation constraints.

Agree the target range, reference uncertainty and acceptance limits before validation.

H04 Council §11, condensed for presentation; S02 p.13; acceptance criteria require the customer

More answers improve completeness.

The Coach shows what changed; confidence still reflects physical constraints.



INFORMATION COMPLETENESS

2 / 10 → 9 / 10

CONFIDENCE

LOW → LOW

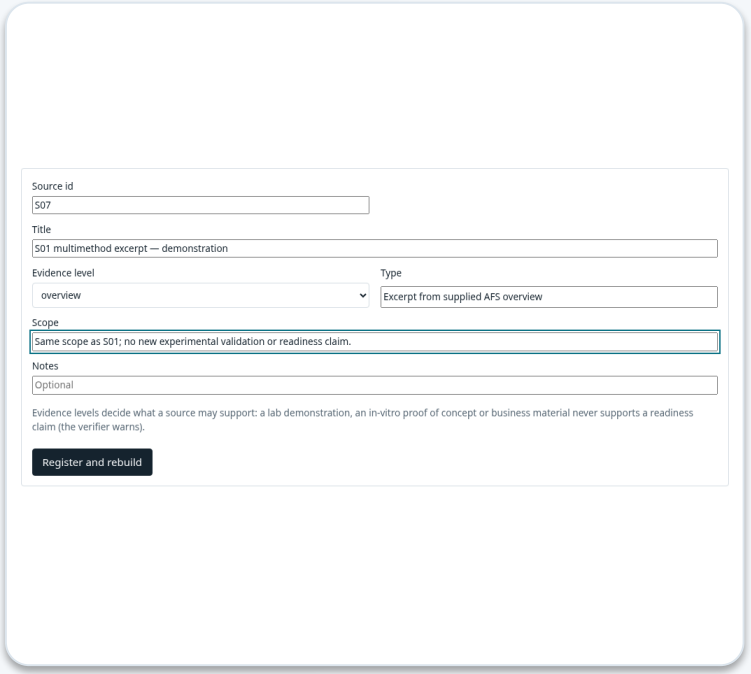
The silica/MIR blocker remains.

The parent run and change summary remain visible.

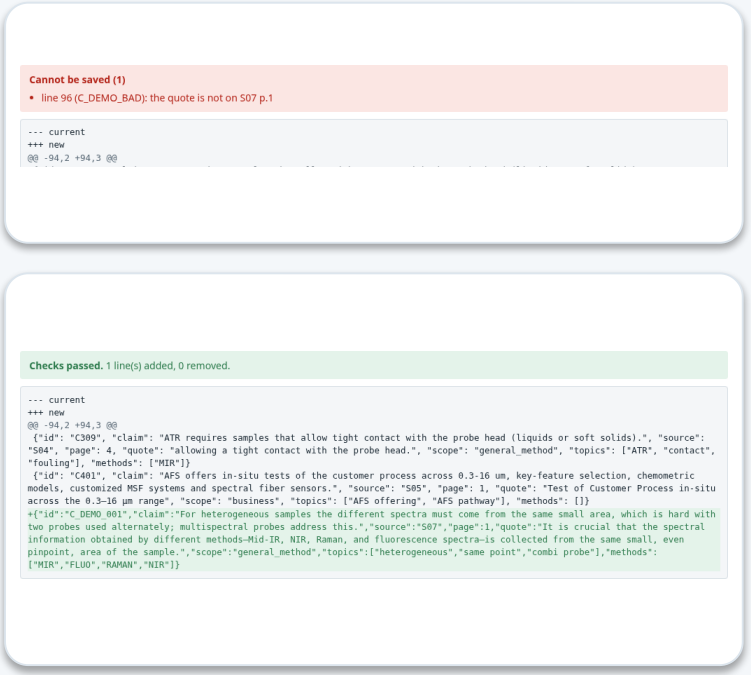
New knowledge must survive a source check.

Register → validate the quote → save a version → rebuild the index

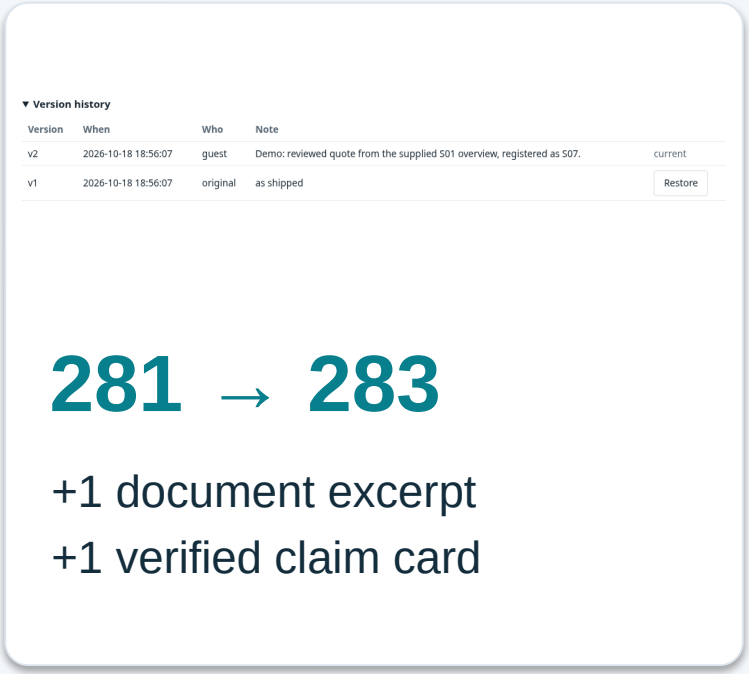
01 Upload + scope



02 Check + diff



03 Version + rebuild



281 → 283

+1 document excerpt

+1 verified claim card

The excerpt reuses supplied S01 evidence. Expansion does not create new scientific validation.

Three priorities before stronger performance claims.

The prototype exposes failure signals; some checks still miss or misclassify claims.

Ground every section

H02 + H05: 4 number flags.

H04: S11 absent from registry.

An R^2 threshold escapes the number counter.

Tighten decision gates

H05: undefined target still provisional.

Negation and units can trigger false warnings.

Keywords can overreach.

Expert validation

No spectra or prediction labels.

No proven detection limit.

Council traces: 72–189 s; use a prepared run for the demo.

Fresh regression evidence: 21 runs completed · 7/7 adversarial expectations · 81 tests passed

Rule-only / hashing baseline; 20 PostgreSQL tests skipped. These are software checks, not Council accuracy.

Make the next experiment reviewable.

01

Clarify

What is actually required?

02

Inspect

Which source supports it?

03

Test

What experiment resolves it?

Working prototype. Inspectable results. Clear validation gaps.

Eisa Atif · Marie Habermann · Atfan Deshmukh · Akshdeep Singh

Naumanurrahman Shaikmujeeburrahman

Different checks answer different questions.

Keep mechanical regression checks separate from stored Council behavior.

Evidence	Conditions	Observed result	What it does not prove
Council archive	6 saved final reports	35/44 checks; 4 number flags	Not fresh; not expert accuracy
Fresh pipeline	21 cases; rule-only / hashing	21 completed; 7/7 adversarial	Not a new LLM benchmark
Verifier totals	Same 21 offline cases	179/179 counted checks	Not semantic correctness
Software suite	SQLite; supplied corpus	81 pass; 20 PG skips	No PostgreSQL claim
Feature demos	Real local UI + API	Coach 2→9; KB +2 records	Illustrative customer answers

Code: 40dd40b · Council archive: 807609d · No case-specific code changes
Mock evaluation: LLM_MOCK=1 · EMBED_BACKEND=hashing · DB_BACKEND=sqlite

Evidence scope travels with the claim.

Sources define what may be concluded, not just what may be cited.

Source	Evidence boundary	Used pages / form
S01 · Multispectral Laboratory	Overview; method selection and sample optics	pp.5–10, 16–18
S02 · Process-Control slides	Company slides; fibers, lab petroleum, solution paths	pp.6, 9, 13, 20
S03 · LED kidney study	In-vitro proof of concept; not clinical readiness	pp.1, 14
S04 · MIR + fluorescence	Laboratory/model-tissue demonstration	6-page paper
S05 / S06 · flyer + link	Business context; not performance validation	Flyer / lab link

Basis: README + frontend integration + pipeline/UI/KB/evaluation code; supplied drafts; official rubric.
Saved Council answers are preserved unedited. Presentation examples are labelled when condensed.